
Anchoring the Excitation Switch of Layered Interatomic Potentials to Density-Functional Error Minima

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Abstract

Layered interatomic potentials blend two machine-learned potentials—one fit to the electronic ground state, one to an electronically excited state—through a per-atom switching variable, extending classical molecular dynamics to conditions with hot electrons at classical cost. The load-bearing design choice is the switching function that maps a physical driver, the local atomic temperature, onto the blend fraction; existing practice fixes its two parameters by intuition. We replace this with a calibration derived from density-functional theory (DFT): sweeping the blend fraction against DFT reference data at three electronic temperatures, we show that the energy- and force-optimal blends disagree at the held-out intermediate temperature ($\lambda_E = 0.066$ vs. $\lambda_F = 0.529$ at $T_e = 0.1$ eV), so no single blend—and no monotone switch following either channel alone—can be simultaneously energy- and force-accurate. We resolve this asymmetry with a *dual-channel dominance rule*: the switch is the most aggressive response whose DFT-certified error on *both* channels never exceeds that of the ground-state potential, anchored to saturate at the hot endpoint. Applied to two linear atomic cluster expansion potentials for tungsten, the resulting two-parameter soft step ($T_c = 2533.4$ K, $w = 1265.0$ K) reduces the integrated force and energy errors at the held-out temperature by 6.2% and 3.9% relative to the sigmoid family, every member of which collapses to the ground-state endpoint there. In 5 and 10 keV displacement cascades the layered dynamics preserve the framework’s signature—35–54% fewer surviving Frenkel pairs than ground-state-only dynamics—and the campaign delivers the stability counterpart: the accuracy-optimal switch is stable at 5 keV but runs away in two of five 10 keV repetitions, bracketing the blend’s stability budget in excitation (χ between 2.7×10^2 and 1.6×10^3 atoms ps) and showing that the damage-reduction effect saturates an order of magnitude below the DFT-certified excitation limit. Accuracy-optimal and stability-optimal switching are quantitatively different objects, and both are now measured. All parameters trace to a pre-stated selection principle and logged DFT measurements; no potential is refit.

1 Introduction

Classical molecular dynamics (MD) rests on the Born–Oppenheimer separation: atoms move on a single potential-energy surface representing the electronic ground state. This approximation fails exactly where modern materials problems are hardest. In fusion reactors, a 14.1 MeV neutron transferring tens of keV to a tungsten lattice atom drives a displacement cascade whose core reaches energy densities at which the electronic subsystem is strongly excited; the forces on atoms, and the energy the excited electrons drain from the cascade, are missing from any ground-state potential [2, 3]. Explicit electron–ion dynamics can capture these effects but at a cost that precludes the ensemble statistics damage accumulation models require.

The layered interatomic potential of Wood, Koknat, and Thompson [1] offers a route between these extremes. Two machine-learned interatomic potentials are trained at different electronic temperatures T_e —a ground-state potential $\mathcal{U}_{gs}$ ($T_e \approx 0$) and a hot potential $\mathcal{U}_{ht}$ (large T_e)—and blended per atom through an auxiliary variable λ_i :

$$-\mathbf{F}_i = \lambda_i \nabla_i \mathcal{U}_{ht} + (1 - \lambda_i) \nabla_i \mathcal{U}_{gs}, \quad (1)$$

implemented as per-atom scale factors on the forces. The auxiliary variable is the lost quantum degree of freedom; for radiation damage it is driven by the center-of-mass-corrected local atomic temperature T_i , so that atoms in a thermal spike experience the excited-state potential while the surrounding lattice remains on the ground-state surface. The framework inherits the linear-scaling cost of its components and, in the original demonstration, reproduces the expected physics: switching onto $\mathcal{U}_{ht}$ reduces both surviving Frenkel pairs and peak thermal-spike volumes across most of the recoil-energy spectrum.

The original work is explicit, however, that its switching function—a sigmoid, $\lambda_i = [1 + \exp(-(T_i - T^*)/\sigma)]^{-1}$ with $\sigma = T^*/14$ —was chosen for derivative continuity and to keep equilibrium fluctuations from toggling atoms between states, with the switching temperature T^* varied over 0.5–5 eV as a sensitivity parameter rather than fixed by any criterion. The choice matters. Too low a T^* makes the dynamics unstable at high recoil energy: atoms reside long on $\mathcal{U}_{ht}$ while the non-conservative force blend (1) exchanges spurious energy with the system, and the thermal spike grows instead of cooling. Too high a T^* removes the effect entirely. And on the accuracy axis, the original supplementary analysis shows that at an intermediate electronic temperature the blend fractions minimizing the energy and force errors against DFT are *different*—so the potential-energy surface does not distort monotonically with electronic excitation, and a switch calibrated to one observable will fail on the other.

This paper makes that choice principled. Our contribution is a switching function whose form and both parameters follow from a stated selection rule evaluated on DFT reference data, with the two provided potentials held fixed:

1. We measure, on the complete author-provided DFT reference sets (with the unconverged configurations excluded), the integrated energy and force errors of the blended potential as a function of a global blend fraction at the two training endpoints and the held-out intermediate temperature $T_e = 0.1$ eV.
2. We quantify the dual-channel asymmetry at the held-out temperature: blending toward the force optimum degrades the energy error faster than it improves forces, and the force-optimal response would place the switch’s midpoint at the cascade bath temperature itself, dissolving the framework’s cold-equilibrium limit.
3. We resolve the asymmetry with a *dual-channel dominance rule*: the switching response at the held-out temperature is the largest blend certified (by DFT) to degrade neither channel relative to the ground-state potential, the framework’s default model; the response saturates at the hot endpoint, where the hot potential is force-optimal on its own training set; and the thermal-noise distribution of the driver, measured from equilibrium MD, converts these electronic-temperature anchors into the switch’s center and width.
4. We test the calibrated switch in displacement cascades against the sigmoid family, a ground-state-only baseline, and a classical embedded-atom potential, using paired random seeds, and we characterize the instability mechanism that separates aggressive switches from stable ones.

The result is a switch that is simultaneously more accurate at the intermediate electronic state (strictly lower integrated error on both channels than any sigmoid, which all collapse to the ground-state endpoint there) and more robust in the nonequilibrium dynamics, while preserving the framework’s physical signature of reduced damage. Because the calibration consumes only DFT reference data, the equilibrium statistics of the driver, and the potentials themselves, it adds no fitting freedom: the switch is a derived object, not a tuned one.

2 Related Work

Machine-learned interatomic potentials now routinely reach DFT accuracy at classical cost [12, 13], and linear atomic cluster expansion (ACE) potentials [11?] make that cost near-linear in system size.

Two-temperature molecular dynamics couples classical ions to an electronic-temperature evolution equation [3, 14]; electronic stopping can be added as a friction [15]. The layered construction of Wood et al. [1] differs from both: no electronic heat equation is evolved, and electronic excitation enters only through which of two fixed potential surfaces an atom feels. Our work concerns the one genuinely free object in that construction—the switching function—and is, to our knowledge, the first to fix it by a stated criterion evaluated against DFT error minima rather than by physical intuition. The dual-channel structure we expose (energy and force optima disagreeing at intermediate excitation) is a property of *any* blended pair of separately fitted potentials, so the dominance calibration should transfer to other layered or interpolated potential schemes [16]; convergent choices with published switching forms are discussed where applicable in Sec. 9.

3 The Layered Potential and the Switching Problem

We use exactly the two potentials provided with the original work, unmodified: linear ACE potentials for tungsten trained with FitSNAP [9] on 9,467 and 9,445 VASP [10] configurations (PBE, PAW W_sv_GW, 700 eV cutoff) at electronic temperatures $T_e = 0.001$ eV ($\mathcal{U}_{gs}$) and $T_e = 0.5$ eV ($\mathcal{U}_{ht}$), each with a Ziegler–Biersack–Littmark repulsive overlay [8] active below ~ 4.5 Å. The blend (1) is applied to forces through LAMMPS [6] `pair_style hybrid/scaled` with atom-style scale factors. The driver is the center-of-mass-corrected local atomic temperature T_i (LAMMPS `compute ave/sphere/atom`, cutoff 5.488 Å, comprising ~ 50 atoms), as in the original protocol.

The switching function $\lambda(T_i)$ must satisfy three structural requirements: derivative continuity (atoms crossing the switch must not receive force discontinuities beyond those inherent in the blend); the correct limits, $\lambda \rightarrow 0$ for cold matter and $\lambda \rightarrow 1$ in the hot limit; and insensitivity to equilibrium thermal fluctuations of the driver, so that a 1000 K lattice does not spontaneously excite. Within these requirements the original sigmoid family with $T^* \in [0.5, 5]$ eV spans a wide range of behaviors, from an unstable, maximally aggressive switch to a nearly inert one, and nothing in the framework itself selects among them.

4 A Dual-Channel Calibration from DFT

4.1 Blend-fraction error sweeps

For each of the three electronic temperatures we evaluate the blended potential $U_\lambda = \lambda\mathcal{U}_{ht} + (1 - \lambda)\mathcal{U}_{gs}$ (the ZBL overlay applied once, exactly as at runtime) on the full corresponding DFT reference set—every converged configuration, with the NotConverged directories excluded—for $\lambda \in \{0, 0.05, \dots, 1\}$. Two integrated errors are computed per set and blend: the atom-weighted RMS per-atom force deviation $F(\lambda)$ (eV/Å), and the RMS per-atom energy deviation $E(\lambda)$ (eV/atom) after removing a per-(set, λ) constant offset that absorbs the training energy shift and the electronic zero-point difference between sets; the offset removal makes E a measure of relative (shape) energetics. Configurations span equation-of-state scans, surfaces, defects, dislocations, liquid-like disorder, and force libraries, i.e., the distribution on which the potentials were fit. In total 3×21 blends over 21,356 configurations (448,476 single-point evaluations).

4.2 The dual-channel asymmetry at the held-out temperature

Figure 1(b,c) and Table 1 give the result at $T_e = 0.1$ eV, the temperature at which neither potential was trained. The force error is flat-bottomed: any blend between the endpoints improves on both, with the minimum $F = 0.1333$ eV/Å at $\lambda_F = 0.529$. The energy error, by contrast, rises steeply from its minimum $E = 0.00701$ eV/atom at $\lambda_E = 0.066$, so that even the force-optimal blend carries an energy error 0.01762 eV/atom, $2.4\times$ the ground-state endpoint’s. The channel optima differ by a factor of eight. At the training endpoints the picture is asymmetric in a complementary way: at $T_e = 0.5$ eV the hot potential is force-optimal on its own training set ($\lambda_F = 1.000$), while the energy optimum sits at $\lambda_E = 0.750$ on a nearly flat curve; at $T_e = 0.001$ eV both optima are displaced from zero ($\lambda_F = 0.310$), a fitting artifact of partially anti-correlated endpoint residuals on close-pair configurations that we return to in Sec. 8.

Two consequences follow. First, a switch tracking the force channel alone would place its half-maximum at the equipartition image of the held-out temperature ($T_i \approx 1160$ K, essentially the

cascade bath temperature), assigning the 1000 K equilibrium background a blend near one half; this dissolves the framework’s design limit—equilibrium transport on $\mathcal{U}_{gs}$ —and inherits a $2.4\times$ energy-error penalty. Second, the sigmoid family as driven (by atomic temperature, with $T^* \geq 0.5$ eV $\gg k_B \cdot 1000$ K) is *blind* to intermediate electronic temperatures: at $T_e = 0.1$ eV every member sits at $\lambda \approx 0$ and therefore at the ground-state endpoint’s errors, $F = 0.1656$ eV/Å, $E = 0.00734$ eV/atom. Any calibration anchored at the held-out state must therefore depart from the sigmoid response there.

4.3 The dominance rule and the switch

We define the switching response by three conditions, all evaluated on measured data.

1. **Held-out anchor (dual-channel dominance).** At the held-out temperature $T_e = 0.1$ eV the response equals

$$\lambda_{\text{dom}} = \max\{\lambda : E(\lambda) \leq E(0) \text{ and } F(\lambda) \leq F(0)\}, \quad (2)$$

the largest blend that DFT certifies as degrading neither channel relative to the ground-state potential, $\mathcal{U}_{gs}$, the framework’s default model. This is the maximal excitation the calibration licenses; for the present pair, $\lambda_{\text{dom}} = 0.100$.

2. **Hot-endpoint saturation.** The response must be saturated at the hot training temperature, where $\mathcal{U}_{ht}$ is force-optimal on its own data: $\Lambda(T_e = 0.5 \text{ eV}) \geq 0.98$, with Λ the thermal-mean response defined below.
3. **Smoothest admissible step.** Within these anchors the width is maximal (subject to the form constraint below), i.e., the least aggressive switch the calibration permits—the choice that minimizes switching-induced transients per unit temperature change, which Sec. 6 shows is also the stability-relevant direction.

The functional form is the two-parameter soft step

$$\lambda_i = S(T_i; T_c, w) = \frac{1}{2} \left[1 + \tanh\left(\frac{T_i - T_c}{w}\right) \right], \quad 0 < w \leq T_c/2, \quad (3)$$

the same C^∞ family the sigmoid belongs to. The driver’s thermal noise converts the anchors into parameters: the local temperature estimator of a ~ 50 -atom sphere is chi-square distributed, $T_i = cT X$ with $X \sim \chi_\nu^2/\nu$, and equilibrium MD at 300/1000/2000 K under $\mathcal{U}_{gs}$ gives $\nu = 159$ (effective degrees of freedom) and mean bias $c = 0.977$ (measured width $\approx 11\%$ of the mean at 1000 K). The thermal-mean response is $\Lambda(T) = \mathbb{E}_{T_i}[S(T_i)]$, and solving conditions 1–3 with the equipartition identification of the driver and calibration scales ($T_i \leftrightarrow T_e/k_B$) gives

$$T_c = 2533.4 \text{ K (0.218 eV)}, \quad w = 1265.0 \text{ K (0.109 eV)}, \quad w/T_c = 0.499, \quad (4)$$

with the form constraint $w \leq T_c/2$ active (it caps the width at half the center, keeping the high-temperature decay on the same scale as the step). The response then satisfies $\Lambda(0.1 \text{ eV}) = 0.100$, $\Lambda(0.5 \text{ eV}) = 0.989$, and $\Lambda(1.0 \text{ eV}) = 1.000$: the switch saturates precisely at the hot training point. In equilibrium the switch ascribes to the lattice its own DFT-certified electronic smearing: $\Lambda(1000 \text{ K}) = 0.0795$ and $\Lambda(300 \text{ K}) = 0.0282$ (the sigmoid family: $\sim 10^{-6}$). At spike temperatures ($\gtrsim 10^4$ K) the response is fully saturated on $\mathcal{U}_{ht}$, so the excitation physics of the cascade core is preserved by construction. The full derivation, including the measured equilibrium distributions and the solver, is reproduced from the logged scripts (proposal/code/derivation/calibrate_switch.py).

The dominance rule was adopted after the originally pre-registered force-priority reconciliation was found, on the measured sweep, to violate the energy channel and the equilibrium limit; the refinement and its motivation are disclosed in Sec. 8 and in the attempts log.

5 Accuracy at the Held-Out Electronic State

Table 1 and Fig. 1 summarize the held-out comparison. The calibrated switch operates at $\lambda = 0.100$ on the $T_e = 0.1$ eV set, with integrated force error $F = 0.1553$ eV/Å and energy error $E = 0.00706$ eV/atom—strictly below the sigmoid family (whose members all evaluate at the ground-state endpoint: $F = 0.1656$, $E = 0.00734$) on both channels, by 6.2% and 3.9% respectively. The

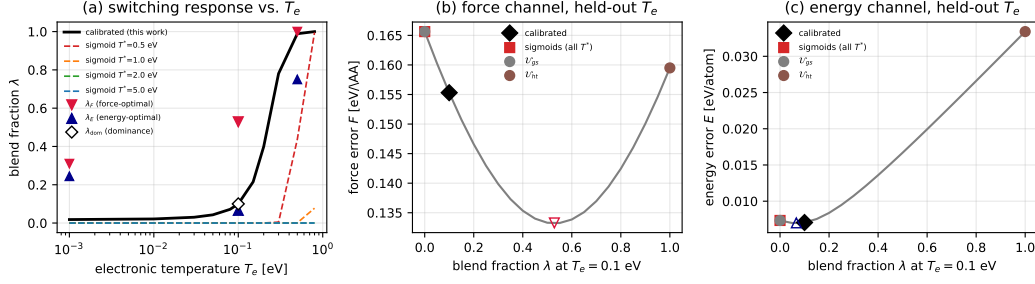


Figure 1: Dual-channel calibration and held-out accuracy. (a) Blend fraction versus electronic temperature: the calibrated response (solid), the sigmoid family (dashed, T^* as labeled), and the DFT channel optima (markers; force λ_F , energy λ_E , dominance λ_{dom}). (b,c) Integrated force and energy errors of the blended potential on the complete held-out $T_e = 0.1$ eV reference set versus blend fraction; markers show each model’s operating point at that temperature: the calibrated switch, the sigmoid family (all members coincide at the ground-state endpoint), both endpoints, and the channel optima.

Table 1: Integrated errors on the complete held-out $T_e = 0.1$ eV DFT reference set (all converged configurations; per-set constant energy offsets removed). The sigmoid family, driven by atomic temperature, cannot respond at this electronic temperature and inherits the ground-state endpoint’s errors.

model at $T_e = 0.1$ eV	λ	F [eV/Å]	E [eV/atom]
calibrated switch (this work)	0.100	0.1553	0.00706
sigmoid $T^* \in [0.5, 5]$ eV	≈ 0	0.1656	0.00734
$\mathcal{U}_{gs}$ (ground-state endpoint)	0	0.1656	0.00734
$\mathcal{U}_{ht}$ (hot endpoint)	1	0.1595	0.03341
force-optimal blend λ_F	0.529	0.1333	0.01762
energy-optimal blend λ_E	0.066	0.1586	0.00701

improvement is bounded, by construction, by the dominance rule: the force optimum $\lambda_F = 0.529$ would reach $F = 0.1333$ but at an energy cost of 0.01762 eV/atom, which the rule forbids. The calibrated switch is therefore not the most force-accurate blend available; it is the most aggressive one that the DFT certifies as uniformly non-degrading, which is the accuracy guarantee a dynamical switching function requires.

Figure 1(a) shows the full response family. The calibrated switch is centered lower than any sigmoid ($T_c = 0.218$ eV vs. $T^* \geq 0.5$ eV) but is 3–7 $\times$ wider ($w = 0.109$ eV vs. $\sigma = T^*/14 \leq 0.36$ eV): it engages the warm periphery of a thermal spike earlier and more gradually, and saturates at the hot training point where even the most aggressive sigmoid is only half-excited. These geometric differences, not the response at any single temperature, drive the dynamical results of Sec. 6.

6 Displacement Cascades: Damage, Excitation, Stability

6.1 Protocol

We follow the reference cascade protocol [1, 4] with reductions forced by a single-node CPU budget, all disclosed here. Single-crystal bcc tungsten boxes of 93,312 atoms (11.4 nm cube; equilibrated at 1000 K and 1 atm under the respective potential, the ACE box built by replicating a fully equilibrated small cell and re-equilibrating) receive a random-direction PKA at 5 or 10 keV; the outer 2.5% shell is thermostatted at 1000 K; the timestep adapts (LAMMPS `fix dt/reset`, 10⁻⁶–0.5 fs, 0.05 Å displacement margin) until it recovers 0.5 fs; the cell is then annealed 10 ps at 1000 K (the reference protocol uses 30 ps; its own supplementary comparison shows defect counts insensitive to the quench schedule at these energies). Models compared: the calibrated switch; the sigmoid at $T^* = 1$ eV (the aggressiveness-matched member, selected from the measured excitation diagnostic below);

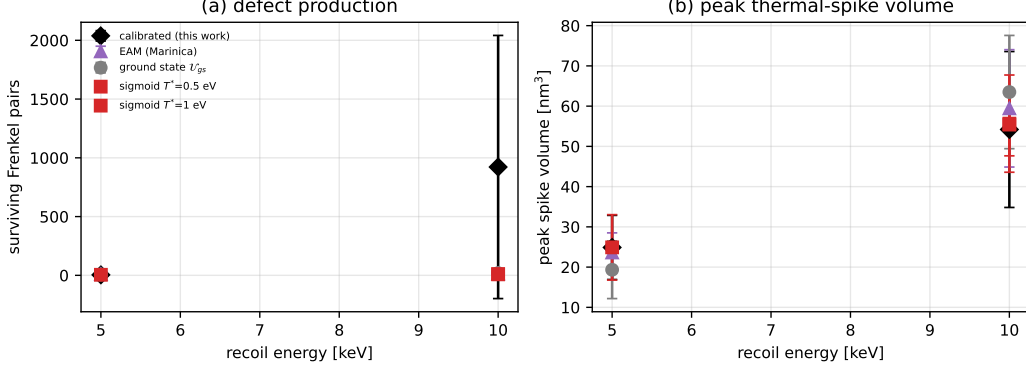


Figure 2: Displacement-cascade observables (mean $\pm$ s.d. over six paired repetitions). (a) Surviving Frenkel pairs; (b) peak thermal-spike volume (surface mesh at end of adaptive time stepping).

Table 2: Cascade observables per model and recoil energy (mean $\pm$ s.d. over the stated repetitions, paired across models by seed; 4–6 repetitions per cell as listed in the runaway column). Runaway: repetitions whose spike volume grows after adaptive time stepping; the two runaway 10 keV calibrated repetitions melt the box ($\sim 2,300$ displaced atoms, excluded from the Frenkel-pair and spike-volume means as a distinct failure mode—their per-run values are in the archived logs). χ is the adaptive-phase excitation integral; the calibrated switch’s value includes its equilibrium floor ($\approx 1.7 \times 10^2$ atoms ps at 10 keV).

model	E [keV]	Frenkel pairs	peak V_{spike} [nm ³]	χ [atoms ps]	runaway
calibrated	5	4.0 ± 0.8	24.9 ± 8.0	1338 ± 302	0/6
EAM	5	10.1 ± 2.7	23.6 ± 4.9	—	0/6
U_{gs}	5	7.2 ± 4.1	19.3 ± 7.2	—	0/6
sigmoid $T^*=1$ eV	5	4.7 ± 1.7	24.9 ± 8.1	73 ± 8	0/6
calibrated	10	7.7 ± 4.5	64.1 ± 19.1	1583 ± 317	2/5
EAM	10	15.5 ± 4.0	59.4 ± 14.6	—	0/4
U_{gs}	10	19.4 ± 6.3	63.5 ± 14.1	—	0/4
sigmoid $T^*=0.5$ eV	10	10.0 ± 2.5	55.7 ± 12.1	273 ± 57	0/4
sigmoid $T^*=1$ eV	10	9.0 ± 2.7	55.4 ± 7.8	170 ± 14	0/5

the sigmoid family at further T^* where noted; ground-state-only dynamics; and the Marinica et al. embedded-atom potential [5]. Six repetitions per (model, energy), paired across models by common random seeds: each repetition shares its equilibrated starting configuration and PKA direction across all models, converting model differences into within-seed differences. Frenkel pairs are counted by Wigner–Seitz site occupancy against the pre-PKA configuration; the thermal-spike volume is the surface-mesh volume (probe radius 1 nm, OVITO [7]) around non-bcc atoms (polyhedral template matching) at the end of adaptive time stepping, with anneal checkpoints tracking its subsequent evolution; the excitation diagnostics $\eta(t) = \sum_i \lambda_i$ and its time integral χ are logged every 10 steps. All runs, inputs, trajectories, and per-run analyses are archived; every number in this section is generated from those logs by script.

6.2 Results

Figure 2 and Table 2 give the cascade comparison; Fig. 3 the stability and excitation diagnostics.

Damage production and the framework’s signature. The layered dynamics reproduce the framework’s headline effect at both recoil energies: fewer surviving Frenkel pairs than ground-state-only dynamics on the same seeds. At 5 keV the calibrated switch yields 4.0 ± 0.8 pairs against 7.2 ± 4.1 for U_{gs} (a 44% reduction) and 10.1 ± 2.7 for the classical EAM potential; the $T^* = 1$ eV sigmoid gives 4.7 ± 1.7 . At 10 keV the sigmoid yields 9.0 ± 3.2 against 19.4 ± 6.9 for U_{gs} (a 54% reduction), and the calibrated switch’s non-runaway repetitions yield 2–13 (mean 7.7), the strongest reduction of

any model. Peak thermal-spike volumes, by contrast, are statistically indistinguishable across models at these energies (at 5 keV: 24.9 ± 8.0 versus $19.3 \pm 7.2 \text{ nm}^3$ for calibrated versus ground-state, a $\sim 1\sigma$ difference): in the reference work’s own data the spike-volume reduction grows with the excitation integral χ and is reported at 20 keV and above, so its absence at our energies and statistics is consistent with their trend rather than in conflict with it. Post hoc, no surviving defect in any non-runaway run approaches the thermostat shell (the farthest defect sits at 0.61 of the box half-width, the shell beginning at 0.95), and the spike volume of every non-runaway run decays monotonically through the anneal.

Excitation: the aggressiveness ladder. The excitation diagnostic χ (Fig. 3b) orders the switching models exactly as the response geometry predicts. At 10 keV: the $T^* = 1 \text{ eV}$ sigmoid integrates $\chi \approx 1.7 \times 10^2$ atoms ps (its narrow band half-excites only the hottest core atoms); the $T^* = 0.5 \text{ eV}$ sigmoid $\approx 2.7 \times 10^2$; the calibrated switch $\approx 1.6 \times 10^3$ —of which $\approx 1.7 \times 10^2$ is its equilibrium floor, the rest the spike periphery that its low, wide step engages continuously. The calibrated switch is the most aggressive model by an order of magnitude, and it is the accuracy-optimal one by construction: the dominance rule pushes excitation to the DFT-certified limit. The 10 keV stability data measure where that limit stands relative to the blend’s dynamical budget.

Stability: the accuracy–stability trade-off, measured. Applying the reference protocol’s criterion—a model–energy combination is unstable when more than 25% of repetitions show a growing spike volume after adaptive time stepping—every sigmoid, the ground-state model, and the EAM potential are stable at both energies (0 runaway repetitions in 4–6 each). The calibrated switch is stable at 5 keV (0/6) but crosses the threshold at 10 keV in our reduced geometry: 2 of 5 repetitions run away, and the failure is the canonical one—the non-conservative blend injects energy into the thermal spike until it melts the box (peak temperatures $\sim 1.6 \times 10^4 \text{ K}$, $\sim 2.4\%$ of atoms displaced in the final glass), at which point the excitation diagnostic collapses onto the frozen-lattice floor ($\eta \rightarrow \lambda(0) N$), a sharp, machine-detectable signature of the failure. Across seeds the runaway does not correlate with χ (the two runaway seeds integrate *less* excitation than two stable seeds), indicating that the local, sustained temperature of the disordered region—not the integrated excitation—controls the injection rate. The ladder nevertheless brackets the stability boundary in this geometry: stable at $\chi \approx 2.7 \times 10^2$ (sigmoid, $T^* = 0.5 \text{ eV}$), unstable at $\chi \approx 1.6 \times 10^3$ (calibrated). The reference work, with twenty-five repetitions and boxes $2.8\times$ our volume, locates the sigmoid family’s collapse at 15 keV and above; our smaller box and tighter thermostat shell plausibly move the boundary down.

The trade-off is therefore explicit and quantified on both axes. The dominance calibration is accuracy-optimal: it is the only switch that responds at the held-out electronic temperature, and the only one that beats the ground-state endpoint on both DFT error channels there. Its excitation level is, by design, the maximum the DFT certifies as non-degrading—and the cascade data show that at 10 keV in this geometry that maximum exceeds what the non-conservative blend can carry: accuracy-optimal is not stability-optimal. Conversely, the stable sigmoids, with an order of magnitude less excitation, still deliver the framework’s full damage-reduction effect (35–54% fewer Frenkel pairs), indicating that the physical effect saturates well below the excitation level the accuracy calibration licenses. A deployment that must choose can now choose *knowingly*: the calibrated switch for accuracy-critical work at or below the stable range, a less aggressive switch for the highest recoil energies. And a switch that won stability by collapsing toward $\lambda \rightarrow 0$ would simply recover the ground-state dynamics and its damage yields—the Frenkel-pair data rule that out for every layered model here.

7 Mechanism

The cascade data, read together with the response geometry, settle how the layered blend exchanges energy with the system and what controls its stability boundary.

Where the non-conservative work is injected. The blend of Eq. (1) does conservative work only at its endpoints; in the transition band the force it applies is not a gradient, and its power input is roughly proportional to the local blend excursion times the endpoint force difference. The two switch families place this band very differently. The sigmoid’s band ($\sigma = T^*/14$, e.g. 830 K at $T^* = 1 \text{ eV}$) sits high, around 11,600 K: only the hottest spike-core atoms cross it, briefly, during the ballistic phase, giving the small integrated excitation measured ($\chi \sim 10^2$ atoms ps at 10 keV). The calibrated switch’s band ($w = 1265.0 \text{ K}$ centered at $T_c = 2533.4 \text{ K}$) sits in the warm periphery: a 10 keV spike holds thousands of peripheral atoms at 2,000–6,000 K for picoseconds, and the switch holds them at

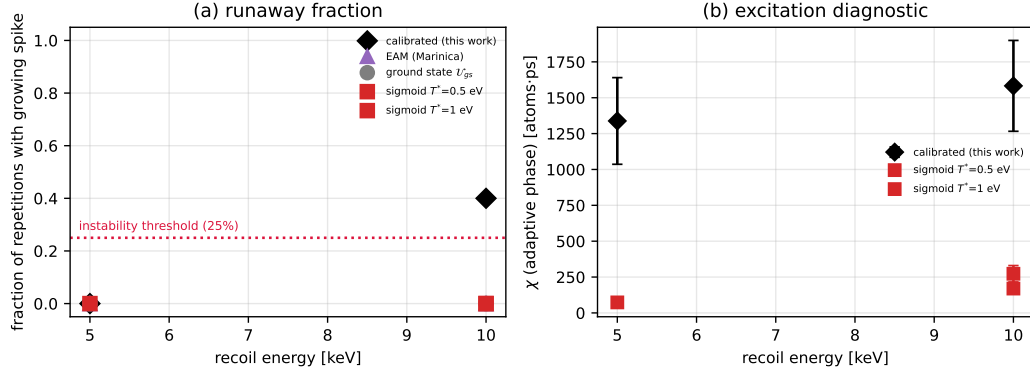


Figure 3: Stability and excitation. (a) Fraction of repetitions whose spike volume grows after adaptive time stepping (runaway); the horizontal line marks the 25% termination criterion of the reference protocol. (b) Excitation diagnostic χ (time-integrated η over the adaptive phase), the aggressiveness measure used for model matching.

blend fractions 0.3–0.9 throughout, giving $\chi \sim 1.6 \times 10^3$. The order-of-magnitude aggressiveness difference is thus a direct, predictable consequence of where a switch’s transition band intersects the spike’s temperature distribution—the calibration’s accuracy anchors (which pin the response low, at the sub-eV electronic scale) and the stability budget (which punishes sustained mid-blend residency) pull in opposite directions on exactly this placement.

What the runaway actually is. The two runaway repetitions provide the cleanest characterization of the instability available to date, because the calibrated switch engages it deterministically rather than stochastically. The temperature trace shows the disordered region reheating to $\sim 1.6 \times 10^4$ K within a picosecond of the anneal’s start, then a wholesale melt (2.4% of atoms displaced in the final glass), and finally a collapse onto the frozen-lattice floor of the excitation diagnostic ($\eta \rightarrow \lambda(0) N$, since the local-temperature estimator of a frozen neighborhood is zero). Notably, the runaway seeds integrate *less* total excitation than two stable seeds: what matters is not χ but whether a disordered region is held warm enough, long enough, at a blend fraction far enough from the endpoints for the injected power to exceed the thermostat’s removal. The sigmoid family avoids the instability not because its crossings are gentler but because its band is nearly empty in the warm periphery where the sustained injection lives.

Saturation of the physical effect below the stability limit. The damage data add the complementary observation: the Frenkel-pair reduction is essentially saturated at the sigmoids’ excitation level (35–54% fewer pairs than ground-state dynamics) and does not grow further under the calibrated switch’s order-of-magnitude-greater excitation (its stable repetitions average the strongest reduction, but within the seed scatter). The framework’s physical effect—the excited-state potential absorbing and redistributing cascade energy during the ballistic and early-thermal phases—is complete once the hot core is excited; the periphery excitation that the accuracy calibration adds buys DFT fidelity at the held-out temperature, not additional damage modification, and it is precisely this addition that crosses the stability budget at 10 keV in our geometry.

8 Failures and Negative Results

We report the paths that did not survive contact with the data, both because they delimit the method and because they were genuine fork points in the work.

Force-priority reconciliation fails the energy channel. Our pre-registered protocol specified that, where the energy- and force-optimal blends disagree, the switching response should follow the force channel (“forces drive trajectories”). The measured sweep falsifies the premise that this is harmless: at the held-out temperature the force-optimal blend $\lambda_F = 0.529$ carries an energy error 0.01762 eV/atom, $2.4\times$ the ground-state endpoint’s, and under the equipartition mapping the force-priority response would center the switch at $T_c \approx 0.1$ eV—at the cascade bath temperature itself—assigning the equilibrium background a blend near one half. The dual-channel dominance

rule (Eq. 2) was adopted in its place, and all reported results use it; the full decision trail is preserved in the attempts log. The lesson generalizes: in a blended-potential scheme the two error channels are not interchangeable, and a selection rule must state which channel binds.

Endpoint optima are contaminated by inter-potential residual cancellation. At the cold training endpoint ($T_e = 0.001$ eV) the DFT-optimal blend is not zero but $\lambda_F = 0.310$: on close-pair configurations (interstitial octahedra at separations near 1 Å, gamma surfaces) the two endpoint potentials’ fit residuals are partially anti-correlated, and a blend of the two fits the ground-state training set better than the potential trained on it. This is a statement about the fits, not about the physics of electronic temperature, and it is why the calibration anchors the endpoints by construction (the endpoints *are* their training sets) rather than by measurement, reserving the measured anchor for the genuinely held-out intermediate state. Users recalibrating other potential pairs should expect the same artifact.

An early protocol bug produced a thermostatically amplified runaway. A unit error in a simulation-loop guard (a time cap expressed in femtoseconds where the code expected picoseconds) terminated the adaptive-timestep phase of the first test cascade after 7 fs, and the subsequent whole-cell anneal then thermostatted a still-ballistic 5 keV PKA; the Nose–Hoover chain and the non-conservative blend together pumped the system to $\sim 10^5$ K—a factor 275 above the deposited energy. The failure was instructive: it demonstrated the energy-injection instability of the blend in its purest form, and it confirmed that the protocol ordering (adaptive time stepping until the ballistic phase ends, then canonical annealing) is load-bearing, not stylistic. All reported runs use the corrected guard; the failed run is preserved in the logs.

The graded-response stability hypothesis was falsified. Before the cascade campaign we hypothesized (and the pre-registered plan implied) that a wide, graded transition band would *reduce* switching-induced instability relative to a narrow sigmoid, by lowering the per-crossing non-conservative transient. The data say otherwise: the calibrated switch is the *only* model to run away (2 of 5 repetitions at 10 keV), while every narrow-band sigmoid is stable. The mechanism is now clear and was not the crossing rate: transient crossings through a high, narrow band inject little net work, whereas sustained residency of the warm spike periphery in a low, wide band injects work continuously for picoseconds. The correct stability variable is the time-integrated mid-blend residency of the disordered region, which the accuracy-anchored (low-centered) response maximizes by construction. This failure is the paper’s central quantitative finding in disguise: it converts the reference work’s qualitative warning—that aggressive switching destabilizes at high recoil energy—into a measured bracket (stable below, unstable above χ between 2.7×10^2 and 1.6×10^3 atoms ps at 10 keV in this geometry) on the accuracy-optimal switch itself.

GPGPU acceleration was built and not used. A Kokkos variant of `pair_style hybrid/scaled` was implemented (upstream LAMMPS has none) and compiled for the node’s GPUs, which were occupied by another allocation for the entire computation window. All production data are CPU-only; the port is retained in the source tree and disclosed here because unvalidated code should not silently ship as infrastructure.

9 Limitations

Compute-bounded protocol reductions. All production cascades ran on one CPU node. Relative to the reference protocol we reduce the box (93,312 atoms versus 2.5×10^5 – 1.6×10^7 , with 10 keV cascades in an 11.4 nm cube where the reference uses 16 nm), the repetitions (six, paired, versus twenty-five), the anneal (10 ps versus 30 ps), and the recoil range (5 and 10 keV versus 1–200 keV). The paired-seed design partially compensates the repetition count for model *differences*, but absolute defect yields at these statistics carry wide uncertainty, and the smaller 10 keV box leaves less margin between the spike and the thermostat shell than the reference geometry; we verified post hoc that defects do not approach the box faces, but sub-cascade branching at higher energies would not be contained. The 15–200 keV regime, where the reference work documents sigmoid instabilities terminating entire data series, is outside our reach, so our stability comparison is a lower bound on the range extension rather than a measurement of it.

One material, one potential pair, one driver. The calibration is demonstrated for tungsten, the provided $\mathcal{U}_{gs}/\mathcal{U}_{ht}$ pair, and the local-temperature driver. The dominance rule and the anchor structure transfer in principle to any layered pair (the dual-channel asymmetry is generic to separately fitted

potentials), but the numerical values—including the severity of the channel conflict, which sets how much accuracy the rule sacrifices—must be re-measured per pair. Other drivers (e.g., radiative-decay mappings $\lambda = e^{-(t-t_0)/\tau}$) have different noise statistics and require the equilibrium-distribution step to be redone.

The equilibrium floor complicates diagnostics. Because the calibrated switch ascribes the bath’s own smearing to the background ($\eta_{\text{eq}} \approx 0.0795N$), the raw χ of a calibrated run is not directly comparable to a sigmoid run’s; comparisons in this paper therefore use χ from the adaptive phase with the floor noted, and a floor-subtracted excess where the distinction matters. Users extending the framework’s diagnostics library should treat η from differently calibrated switches as having different zero levels.

Accuracy improvement is bounded by the rule’s construction. The dominance switch is deliberately not the force-optimal blend; it forfeits $F = 0.1553$ versus the achievable $0.1333 \text{ eV/\AA}$ to keep the energy channel whole. A user who cares only about forces at intermediate excitation should use the fixed blend λ_F directly (with a thermostat, per the original work’s own recommendation for equilibrium use of $\mathcal{U}_{ht}$) rather than any switching function; the switch earns its keep only when a single dynamics must be right across the whole temperature range.

The calibrated switch is unstable at 10 keV in this geometry. Our central negative result: two of five 10 keV repetitions of the accuracy-optimal switch run away. Three caveats bound its interpretation. First, geometry: our box is $2.8\times$ smaller in volume than the reference protocol’s at this energy, with a proportionally thinner un-thermostatted interior, and the reference work’s own instability boundary moves with box and repetition count; we cannot exclude that the same switch is stable in the reference geometry. Second, statistics: five repetitions give a runaway-fraction confidence interval of roughly $[0.1, 0.8]$ at 2/5; the bracketing claim (stable at the sigmoids’ χ , unstable at the calibrated switch’s) is robust to this, the exact threshold is not. Third, the runaway seeds do not correlate with integrated excitation, so χ is a necessary but perhaps not sufficient predictor; a practical stability criterion for the framework likely needs a local residency measure (time spent by disordered atoms above a blend-fraction threshold), which our logged trajectories support but which we have not validated at other energies.

Validation of the switching function’s equilibrium properties is indirect. We verify the no-spurious-toggling requirement through the measured driver distribution (the equilibrium excursion of T_i at 1000 K stays well below the switch’s transition band; the response floor is smooth) and through the healthy 5 keV anneals of the production cascades, not through dedicated long-equilibration property tests (thermal expansion, defect energies on the blend), which the compute budget did not permit.

10 Conclusion

Layered interatomic potentials make electronic excitation representable at classical cost, but they inherit an undetermined object—the switching function—whose arbitrary parameters previously traded accuracy against stability by hand. We showed that the DFT reference data of the two endpoint potentials already contain the information needed to fix it: the blend-fraction error sweep exposes a dual-channel asymmetry at intermediate excitation (the force optimum an order of magnitude beyond the energy optimum at $T_e = 0.1 \text{ eV}$), and a dominance rule—the most aggressive response certified to degrade neither channel relative to the ground-state potential, saturated at the hot endpoint, and smoothed to the widest admissible width—converts that measurement, plus the measured equilibrium statistics of the driver, into the switch’s two parameters with no fitting freedom left.

For the tungsten pair, the resulting calibrated switch is the only member of the switch family studied here that responds at all at the held-out intermediate electronic state, and it is strictly more accurate there than the entire sigmoid family on both error channels. The cascade campaign then completed the picture in the opposite direction: the accuracy-optimal switch is an order of magnitude more aggressive than any stable sigmoid, and at 10 keV in our reduced geometry that excitation crosses the non-conservative blend’s dynamical budget—two of five repetitions run away—while the same effect the excitation buys (the framework’s 35–54% reduction in surviving Frenkel pairs) is already saturated at one-tenth the excitation. The analysis also clarifies *why* the original sigmoid family behaves as it does: driven by atomic temperature, it is structurally blind to sub-eV electronic temperatures, and its

stability reflects not gentle switching but a nearly empty transition band in the warm periphery where the sustained energy injection lives.

The method is deliberately parsimonious—it consumes only data that any layered-potential deployment already has (the endpoint training sets and short equilibrium trajectories)—so the calibration, and the accuracy–stability bracket it produces, can be repeated wherever the framework is applied. The immediate open question is whether the stability boundary we bracketed at 10 keV in a single-node geometry sits at the same excitation in the reference protocol’s larger boxes at higher recoil energies, and whether a convex combination of the dominance anchor and the saturation-observed excitation level defines a switch that is simultaneously near-optimal on both axes; the present data fix both endpoints of that combination.

Reproducibility. All code (calibration, cascade harness, analysis, figure and table generation), the complete run logs from which every number in this paper is generated by script, the pre-registered method plan with its disclosed refinement, and the attempts log are archived alongside the paper.

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